Identifiability of species interactions in a living material: Bayesian calibration of an extended Hamilton principle multiphase model

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Abstract

Biofilms are living materials whose composition, and with it their mechanical response, is set by interactions between species that cannot be measured directly. We treat the identification of these interactions as a model-based inverse characterization problem and calibrate a thermodynamically consistent multiphase model of a five-species oral biofilm, derived from the extended Hamilton principle, against in-vitro time courses under four combinations of community state and cultivation. All 15 entries of the symmetric interaction matrix are inferred jointly by transitional Markov chain Monte Carlo, with the stiff forward model compiled in JAX and evaluated for all particles in parallel on a GPU (∼150× faster than on a CPU). The data enter in stages, from species composition to community viability, and every stage is accepted only after explicit checks of seed agreement, mixing and posterior mass at the prior bounds. The calibrated model reproduces the species fractions with root-mean-square errors of 0.02–0.09 in three conditions[**prov.**]; in the commensal static condition (0.17–0.18) the composition and the measured viability cannot both be matched. Only part of the interaction matrix is determined by the data—four and six entries in the two commensal conditions, all among the early colonisers and with the same signs where they over-

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lap, and eight and nine in the dysbiotic ones[prov.]—and we state explicitly which. The coefficient between *F. nucleatum* and *P. gingivalis*, positive in the dysbiotic flow-reactor community when only the composition is used, is no longer sign-determined once the viability is added[prov.]. The measured pH, not used in the inference, follows the predicted composition through a fixed regression in that community (with an offset of about 0.13 in its commensal counterpart), and removing *F. nucleatum* from the calibrated model abolishes the late increase of *P. gingivalis*, a prediction open to direct experimental test[prov.].

Keywords: Inverse characterization, Bayesian calibration, Uncertainty quantification, Extended Hamilton principle, Living materials, GPU

1. Introduction

Biofilms are soft, living materials: microbial cells embedded in a self-produced matrix whose composition evolves in time and determines their mechanical response [1]. On dental implants, a shift in the species composition of the biofilm from a commensal towards a dysbiotic community is associated with peri-implantitis [2, 3, 4]. Which interactions between the species drive such a shift is difficult to measure: co-culture experiments report how the composition changes over time, not the interactions themselves.

Continuum mechanics offers a route from composition to interaction. Based on the extended Hamilton principle [5], Klempt et al. [6] derived a multiphase model in which species volume fractions and viabilities evolve under a free energy that contains a symmetric interaction matrix $\mathbf{A}$; thermodynamic consistency and the symmetry of $\mathbf{A}$ follow from the variational structure. Calibrating $\mathbf{A}$ against data is an inverse characterization problem: the physics fixes the structure of the model, and the data determine its coefficients. Data-driven methods for such problems are most reliable when they are guided by physics and by explicit checks of what the data can determine. Once calibrated, the model can be used as an in-silico test bed, for example to ask how the community would develop if one species were removed. Bayesian updating has been used to calibrate constitutive parameters of biofilm models with surrogates [7], but the interaction matrix poses two specific difficulties. First, with five usable time points and five species, at most 25 observations constrain 15 unknowns, so identifiability cannot be assumed. Second, the coupled volume-fraction and viability dynamics pro-

duce posteriors that are multimodal and bounded by the prior box, for which the convergence of standard samplers is hard to verify.

Here we infer all 15 entries of $\mathbf{A}$ from the time courses of Heine et al. [8] for five oral taxa—*Streptococcus oralis* (So), *Actinomyces naeslundii* (An), *Veillonella* spp. (Vei), *Fusobacterium nucleatum* (Fn) and *Porphyromonas gingivalis* (Pg)—in four conditions: a commensal or a dysbiotic community, each under static cultivation in well plates or dynamic cultivation in a flow reactor (HOBIC, Hannoverian Oral multispecies Biofilm Implant flow Chamber), abbreviated CS, CH, DS and DH (commensal/dysbiotic, static/HOBIC). We combine three elements: first, transitional Markov chain Monte Carlo (TMCMC) [9, 10] with the forward model compiled in JAX [11] and evaluated for all particles in parallel on a GPU; second, a staged introduction of the data, from species composition to community viability; and third, explicit acceptance criteria for every stage, including a rule for widening prior bounds at which posterior mass accumulates, and an analysis of which interactions the data determine. The data determine only a subset of $\mathbf{A}$, which we identify entry by entry; where the commensal conditions overlap, the identified entries agree in sign across cultivation modes, whereas the Vei–Pg coefficient of the dysbiotic community changes sign between static and dynamic cultivation[**prov.**].

2. Model

At a material point, species $i = 1, \dots, 5$ occupies the volume fraction $\phi_i \in (0, 1)$, of which the fraction $\psi_i \in (0, 1)$ is alive; the living volume fraction is $\bar{\phi}_i = \phi_i \psi_i$. A void fraction ϕ_0 closes the volume, $\sum_{l=0}^{5} \phi_l = 1$. As in Klempt et al. [6], the evolution equations result from the stationarity of the rate functional (derivation in Supplementary Section S1)

$$\dot{\Psi} + \Delta + C \to \mathrm{stat}_{(\dot{\boldsymbol{\phi}}, \dot{\boldsymbol{\psi}}, \gamma)}, \tag{1}$$

the local form of the extended Hamilton principle [5, 12, 13, 14], with the free energy, dissipation and constraint

$$\Psi = -\tfrac{1}{2}\, c^* \, \bar{\boldsymbol{\phi}}^{\mathsf{T}} \mathbf{A}\, \bar{\boldsymbol{\phi}}, \qquad \Delta = \tfrac{1}{2}\, \dot{\bar{\boldsymbol{\phi}}}^{\mathsf{T}} \boldsymbol{\eta}\, \dot{\bar{\boldsymbol{\phi}}} + \tfrac{1}{2}\, \dot{\boldsymbol{\phi}}^{\mathsf{T}} \boldsymbol{\eta}\, \dot{\boldsymbol{\phi}}, \qquad C = \gamma \Big(\sum_{l=0}^{5} \phi_l - 1 \Big). \tag{2}$$

Here c^* is the nutrient concentration, $\boldsymbol{\eta} = \mathrm{diag}(\eta_i)$ with $\eta_i = 1$, and γ is a Lagrange multiplier; the antibiotic term of the original model vanishes

because the experiments contain no antibiotic. For $n = 5$ species, stationarity gives $2n + 2 = 12$ coupled non-linear equations, for $i = 1, \ldots, 5$,

$$0 = -c^* \psi_i (\mathbf{A}\bar{\boldsymbol{\phi}})_i + \eta_i \left(\dot{\phi}_i \psi_i^2 + \bar{\phi}_i \dot{\psi}_i + \dot{\phi}_i \right) + \gamma, \tag{3}$$

$$0 = -c^* \phi_i (\mathbf{A}\bar{\boldsymbol{\phi}})_i + \eta_i \left(\dot{\psi}_i \phi_i^2 + \bar{\phi}_i \dot{\phi}_i \right) + \gamma, \tag{4}$$

together with $0 = \gamma + \dot{\phi}_0$ and the volume constraint. They are integrated by an implicit Euler scheme with Newton iterations (2500 steps over the 21-day window), with a logarithmic barrier keeping ϕ_i, ψ_i inside $(0, 1)$. Only the symmetric part of $\mathbf{A}$ enters the quadratic free energy, so $\mathbf{A}$ is taken symmetric, with 15 independent entries, which form the parameter vector $\boldsymbol{\theta} \in \mathbb{R}^{15}$. The coefficient a_{ij} is an effective, undirected interaction of the model; it lumps mechanisms that the model does not resolve, such as metabolite exchange or co-aggregation. Only the product $c^* a_{ij}$ enters, so we fix $c^* = 25$ to keep $a_{ij} = \mathcal{O}(1)$. Removing a species is represented by holding its volume fraction at zero.

3. Bayesian inference

Likelihood.. The data are the species fractions at days 3, 6, 10, 15 and 21 (day 1 is the initial condition), the medians over eight biological replicates [8]. Model time is the time since inoculation, $t = s\,d$ with $s = 0.2375/21$; the day-1 composition, the earliest measurement, serves as the initial state at $t = 0$. In the flow reactor the inoculated medium is replaced by sterile medium after 24 h [8], so the growth phase there can also be taken to start at day 1. Starting the model at day 1 ($t = s\,(d-1)$) changes the composition error of the maximum a posteriori estimate (MAP) by at most 0.008 in any condition without re-inference[**prov.**]. The model fractions are renormalised over the five species, $\hat{y}_i = \phi_i / \sum_j \phi_j$, and compared through a weighted Gaussian log-likelihood

$$\ln p(\mathbf{y}_{\mathrm{obs}} \mid \boldsymbol{\theta}) = -\frac{1}{2} \sum_k \sum_{i=1}^{5} w_{ik} \, \frac{\left(y_{\mathrm{obs},i}(t_k) - \hat{y}_i(t_k; \boldsymbol{\theta}) \right)^2}{\sigma_i^2} + \mathrm{const}, \tag{5}$$

with σ_i the replicate standard deviation of species i. The weights up-weight Pg ($\lambda_{\mathrm{Pg}} = 5$), whose late increase is the main dynamic feature of the dysbiotic data, and the last two time points ($\lambda_{\mathrm{late}} = 3$), and down-weight species with a mean fraction below 5% ($\lambda_{\mathrm{rare}} = 0.1$; Supplementary Table S2); without

these weights the median of a_{45} in DH is unchanged but its interval widens (Supplementary Section S9). In the later stages a second Gaussian channel compares the predicted community viability $\sum_i \bar{\phi}_i / \sum_i \phi_i$ with the measured fraction of membrane-intact cells, weighted by $\lambda_\psi = 1.5$ in the commensal conditions, 2.0 in DS and 3.0 in DH, where the viability of Pg carries most of the signal (Supplementary Table S2). The measured pH under dynamic cultivation is not used in the inference and serves as a check (Section 4).

Prior and stages.. The prior is uniform on a box $[l_k, u_k]$ for each entry, set to contain earlier estimates; it encodes no sign constraint. Inference proceeds in four stages on the same data (Fig. 1): (i) composition likelihood on the box; (ii) as (i), on the box narrowed to the stage-(i) posterior mean ± 4 standard deviations; (iii) composition and viability, back on the full box, starting from the stage-(ii) particles; (iv) as (iii), on the box narrowed to the stage-(iii) posterior mean ± 4 standard deviations. A ± 4 SD box removes less than 10^{-4} of a Gaussian marginal, so narrowing shortens the tempering without truncating the posterior. If more than 20% of the posterior mass of an entry lies in the outer 5% of the box on one side, the bound is moved outwards and the stage is repeated; the boxes finally used are listed in Supplementary Table S4.

TMCMC on the GPU.. TMCMC [9, 15] passes a particle population through the tempered posteriors $p_m(\boldsymbol{\theta}) \propto p(\mathbf{y}_{\mathrm{obs}} \mid \boldsymbol{\theta})^{\beta_m} p(\boldsymbol{\theta})$, $0 = \beta_0 < \cdots < \beta_M = 1$, choosing each increment so that the coefficient of variation of the importance weights stays at a target value [10]. After resampling, every particle performs K Metropolis–Hastings steps with random-walk proposals scaled by the weighted particle covariance, alternating with differential-evolution proposals [16] that follow strongly correlated directions; proposals outside the box are rejected without solving the model. Each likelihood evaluation requires one solution of Eqs. (3)–(4), and a run requires 10^5–10^7 of them. We implement the implicit solver, including Jacobian assembly and linear solve, in JAX, compile it into a single XLA kernel and evaluate all particles at once with `vmap`. On one GPU (NVIDIA RTX 4090) this is 170 times faster per tempering step than a 12-core CPU at matched settings (147 times end to end; Supplementary Table S3), which makes it affordable to repeat every stage with three seeds.

Acceptance criteria.. Every stage is run with three independent seeds and accepted only if (a) the stored log-likelihood of every particle agrees with a

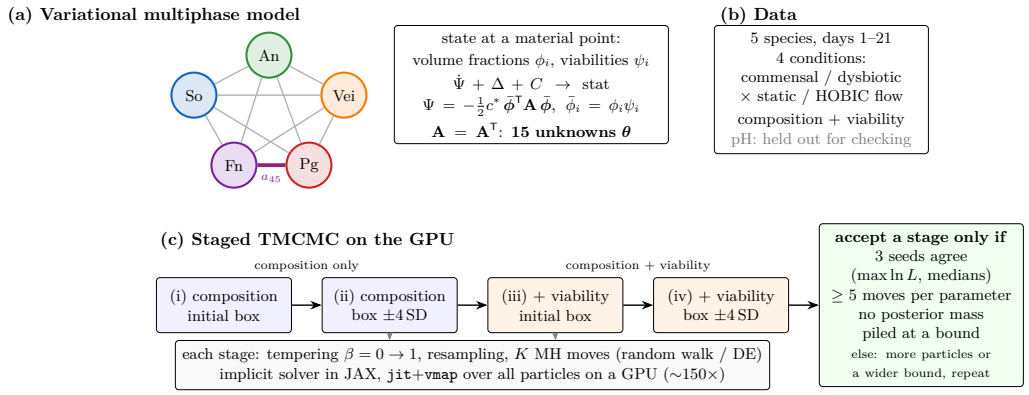


Figure 1: Model and inference. A material point carries the volume fractions ϕ_i and viabilities ψ_i of five taxa, coupled through the symmetric interaction matrix $\mathbf{A}$ (15 unknowns). TMCMC with a GPU-compiled forward model infers $\mathbf{A}$ in four stages, from species composition to community viability; every stage is accepted only after the checks (a)–(e).

recomputation; (b) the tempering reaches $\beta = 1$; (c) each particle is moved at least twice per tempering step and at least five times per free parameter over the run; (d) the maximum log-likelihood differs by at most 1 across seeds; and (e) every posterior median differs across seeds by at most half a posterior standard deviation; an entry whose pooled posterior standard deviation is at least 0.8 times that of a uniform distribution on its box is reported as not identified by the data and excluded from this criterion, and for an entry with several posterior modes the criterion is applied within each mode, with the mode weights reported together with their spread across seeds. Criterion (c) counts accepted moves rather than the acceptance rate: with optimally scaled random-walk proposals [17], $5d$ accepted moves displace each coordinate by about five posterior standard deviations, independently of the dimension d. Stages that fail are repeated with more particles or mutation steps, never interpreted. The evidence estimate of TMCMC is used only to compare models on the same data and box (Section 4).

4. Results and discussion

4.1. Fit and convergence

All stages on the path to the final posterior were accepted under criteria (a)–(e) with three seeds each; the stage-(i) runs used only for the Bayes fac-

tors do not all meet (c) and (e) (Supplementary Table S5) **[DS, DH stage (iv) pending; their numbers below are from stage (iii)]**. At stage (iv) the composition is reproduced with a root-mean-square error of 0.076–0.078 in CH, 0.172–0.179 in CS, 0.021–0.028 in DS and 0.081–0.090 in DH**[prov.]** (Fig. 2). Accepting a stage often required more than one attempt: posterior mass accumulated at a bound of the initial box in every condition and was released by widening that bound (Supplementary Table S4), and the dysbiotic conditions needed up to **[18000 (DS) and 20000 (DH)]** particles before the three seeds agreed (Supplementary Table S5). Without the acceptance criteria, several of these runs would have been read as converged.

4.2. What the data determine

Figure 3 and Supplementary Table S6 classify every entry of $\mathbf{A}$ at stage (iv); the number of directions in which the likelihood itself is narrow at the MAP is 4, 4, 8 and 5, against 4, 6, 8 and 9 identified entries (Supplementary Section S6). In the commensal conditions only a few entries are identified: four in CS (a_{11}, a_{22}, a_{33}, a_{13}) and six in CH (a_{11}, a_{12}, a_{22}, a_{33}, a_{13}, a_{23}), all among the early colonisers So, An and Vei. Where both conditions identify an entry, the sign agrees: the self-interaction of So is negative (a_{11}, 90% intervals $[-4.9, -1.2]$ in CS and $[-2.5, -0.5]$ in CH) and the So–Vei coefficient is positive (a_{13}, $[+2.8, +9.2]$ and $[+0.01, +1.4]$), in line with the lactate cross-feeding between streptococci and *Veillonella* [2], although a_{13} is an effective coefficient and not a measurement of that mechanism. Two further entries of CH, a_{14} and a_{15}, are determined in sign only (90% intervals $[-4.8, -0.2]$ and $[-4.8, -0.3]$), and every other entry involving Fn or Pg spreads over its whole box. This is expected: Fn and Pg stay below 1% of the commensal communities from day 3 on, so the composition data carry little information on their interactions. In the dysbiotic conditions, where Fn and Pg reach 10–25% of the community, more entries are identified: eight in DS and nine in DH, now including interactions of the late colonisers**[prov.]**. The Vei–Pg coefficient a_{35} is identified in both but with opposite signs, positive under static cultivation (DS, $[+0.3, +2.4]$) and negative in the flow reactor (DH, $[-13.4, -1.7]$), and the An–Fn coefficient a_{24} is positive in DH ($[+2.7, +7.1]$). A sign change between cultivation modes is a statement about the effective model under different nutrient and flow conditions, not about the pairwise mechanism itself. Comparisons between the commensal and the dysbiotic community are confounded by design: the two communities differ not only in their ecological state but also in the *Veillonella* species (*V. dispar* vs. *V. parvula*) and the

Pg strain (DSM 20709 vs. W83) [8], so differences between their matrices combine state, strain physiology and cultivation; the DS–DH contrast above compares the same strains and is free of this confound.

The classification is a result in its own right: it states which conclusions about the interaction network the data can support, and it is obtained from the same runs at no additional cost.

4.3. The Fn–Pg coefficient

The coefficient a_{45} links the bridging organism Fn to the pathogen Pg. Comparing the full model with a model in which $a_{45} = 0$, both inferred at stage (i), gives $\ln B = +0.10$ (CS), $+0.05$ (CH), -2.67 (DS) and $+1.94$ (DH) (Supplementary Table S7). Because the uniform prior penalises the full model by the log-width of the a_{45} box, $\ln B$ is read only within a condition: in DS, where the box had to be widened to $[-15, 20]$, the penalty alone is $\ln 35 \approx 3.6$. In DH, the data favour a non-zero coefficient (Supplementary Table S7). With the composition alone (stage (i)) a_{45} is positive in all seeds (90% interval about $[+0.5, +5.5]$); without the likelihood weights its median is unchanged but the lower 5% bound drops to -0.06 to $+0.17$ across seeds (Supplementary Section S9). Once the viability enters, the median of a_{45} stays positive in both dysbiotic conditions ($+2.8$ in DS, $+1.2$ in DH) but the 90% intervals include zero ($[-1.3, +15.1]$ and $[-1.9, +7.5]$)**[prov.]**. The composition data alone thus support a positive Fn–Pg coefficient in DH, and the viability data neither confirm nor contradict it.

4.4. Checks against data not used in the inference

The pH measured under dynamic cultivation was not used in the inference (Fig. 5b,c). A linear regression of the measured pH on the measured species fractions (intercept and four fractions, Vei as reference; $R^2 = 0.93$ over 90 time points in CH and DH) was fitted once on the data alone and then applied, unchanged, to the posterior predictive composition. In DH the predicted pH follows the measured rise from about 6.3 to 7.0 ($R^2 = 0.63$, RMSE 0.12)**[prov.]**; in CH it is offset upwards by about 0.13, because the model keeps Fn and Pg near 4–5% where the data show below 1%, a consequence of down-weighting rare species in the likelihood, and these two taxa carry the largest regression coefficients. The calibrated model can also be used in silico: removing Fn from the dysbiotic flow-reactor community abolishes the late increase of Pg (Fig. 5a). The day-21/day-15 ratio of Pg falls from 2.9 (90% interval $[2.3, 3.5]$; measured 2.74) to 1.1 ($[0.3, 1.8]$), and only

2–7% of the posterior samples keep a ratio of 1.5 or more[**prov.**]. Setting only $a_{45} = 0$ removes the increase in most but not all samples (16–52% keep it, depending on the seed), so the prediction rests mainly, though not only, on the Fn–Pg coefficient; it can be tested directly by a co-culture without Fn. Under static cultivation the model shows no comparable late increase (ratio 1.1), so the knockout is not informative there.

4.5. Limitations

The likelihood treats the five species fractions as independent although they sum to one, so the posterior intervals may be somewhat too narrow. The fit is poorest in CS, almost entirely in So and Vei (per-species errors 0.31 and 0.24, all others below 0.05), where the So fraction rises from 0.18 to 0.70 between day 1 and day 3. The model can follow this rise: fitted to the composition alone it reaches an error of 0.075, but only with a live fraction near 0.97, against the measured 0.65–0.77. The composition and the live/dead staining cannot both be matched in CS, and the posterior settles on a compromise; we report it rather than re-weight the channels for this condition. The compromise is sharpest at the MAP: the bulk of the CS posterior lies closer to the composition data (pointwise posterior median, error 0.125), whereas the MAP, identical in all three seeds, sits where the live fraction is matched best (error 0.179); in the other conditions the two differ by less than 0.02 (Supplementary Section S7). The interaction coefficients are effective quantities of this model and data set; they lump mechanisms that the model does not resolve, such as metabolite exchange and pH, and are not direct measurements of pairwise mechanisms. The calibration covers a 0D material point; the spatial structure of the biofilm and its mechanics are not part of the inference. The calibrated dynamics are, however, the input that continuum growth models of biofilms require [1]: the total volume fraction of the point model can drive the growth variable of a finite-element model, so that differences in growth-induced stress between conditions follow from the calibrated interactions rather than from assumed species-specific growth weights. Because the likelihood uses species fractions only, the absolute amount of biomass is a prediction of the model rather than a calibrated quantity, and its posterior spread has to be propagated into such a coupling.

5. Conclusions

The interaction matrix of a variational multiphase biofilm model was calibrated against time courses of a five-species oral community in four conditions, with a GPU-compiled forward model, a staged likelihood and explicit acceptance criteria for every stage. Three points carry over to other inverse characterizations of living materials. First, convergence has to be checked, not assumed: posterior mass at the prior bounds and disagreement between seeds were found in every condition and removed only by widening bounds and adding particles. Second, identifiability is a result: with at most 25 composition values for 15 unknowns the data determine only some interactions, and stating which ones limits the biological conclusions that can be drawn from the model. Third, a misfit can be informative: in the commensal static condition the composition and the viability data cannot be matched at once, which may point to a process the model lacks, or to a mismatch between the live/dead stain and the model viability, rather than to a failed calibration. In the dysbiotic conditions more interactions are identified, and the Fn–Pg coefficient, which the composition data alone suggest is positive in DH, is not sign-determined by the full data set**[prov.]**.

Declaration of competing interest

The authors declare no competing interests.

Data and code availability

Code, run configurations, convergence records and posterior samples are archived at Zenodo (doi:10.5281/zenodo.23320423). The experimental data are those of Heine et al. [8].

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CRediT authorship contribution statement

[draft, to be confirmed by all co-authors] **Keisuke Nishioka:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. **Felix Klempt:** Methodology, Software, Validation, Writing – review & editing. **Hendrik Geisler:** Validation, Writing – review & editing. **Rumjhum Mukherjee:** Validation, Writing – review & editing. **Nils Heine:** Resources, Data curation, Writing – review & editing. **Katharina Doll-Nikutta:** Resources, Writing – review & editing. **Meike Stiesch:** Resources, Funding acquisition, Writing – review & editing. **Szymon P. Szafrański:** Validation, Writing – review & editing. **Meisam Soleimani:** Conceptualization, Methodology, Supervision, Project administration, Writing – review & editing. **Philipp Junker:** Conceptualization, Supervision, Funding acquisition, Writing – review & editing.

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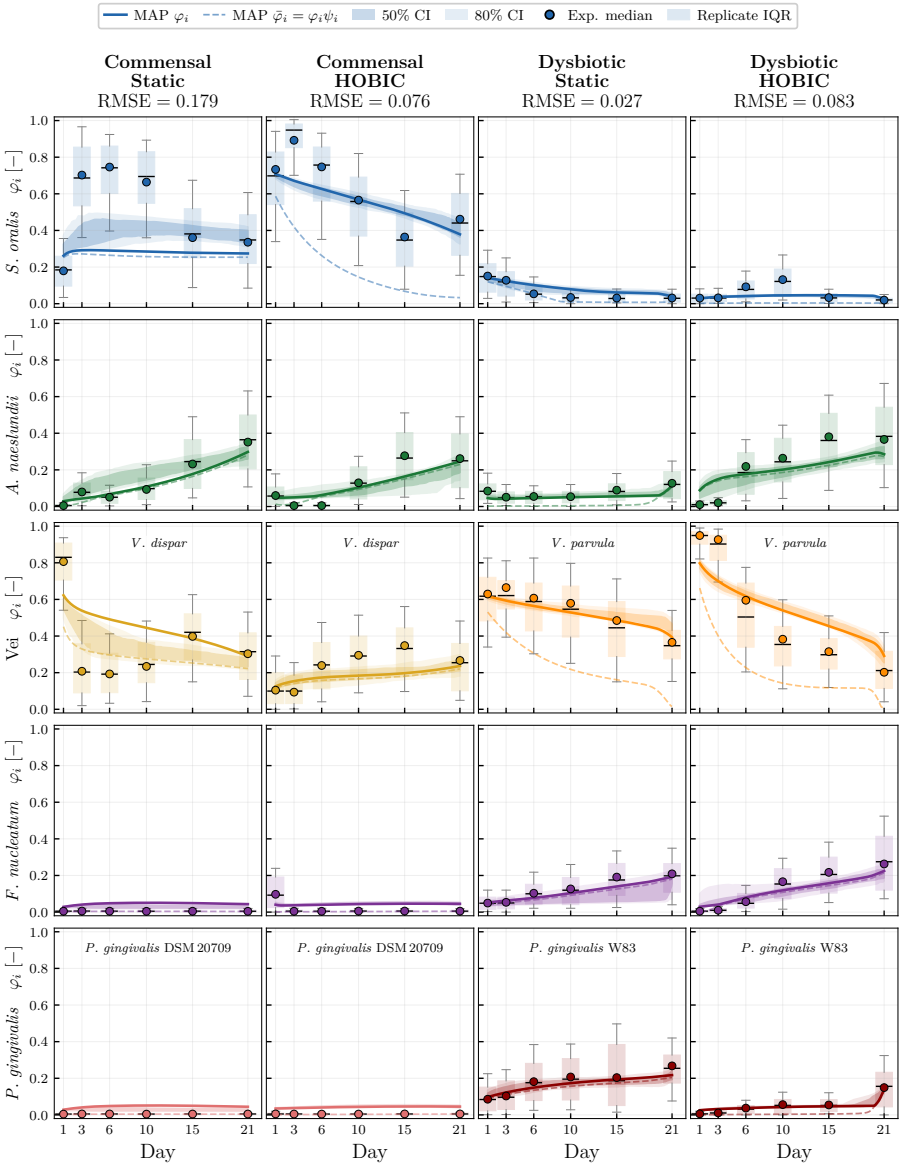


Figure 2: Species fractions over 21 days in the four conditions (columns) for the five taxa (rows). Lines: MAP of stage (iv), species volume fraction ϕ_i (solid) and living fraction $\bar{\phi}_i$ (dashed); bands: 50% and 80% posterior predictive intervals pooled over three seeds; dots and boxes: median and interquartile range of the replicates [8]. The RMSE of the MAP is given above each column. **[DS and DH from stage (iii) until stage (iv) is accepted.]**

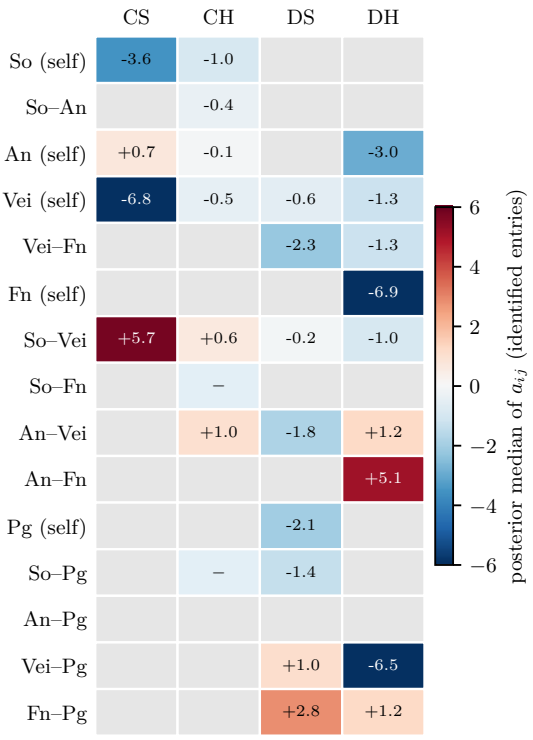


Figure 3: What the data determine. Entries of **A** (rows) in the four conditions (columns) at the final stage, three seeds pooled. Coloured with value: identified (posterior standard deviation below 0.8 times that of a uniform distribution on the box), colour and number give the posterior median; light colour with sign: only the sign is determined (90% interval excludes zero); grey: not identified. **[DS, DH from stage (iii).]**

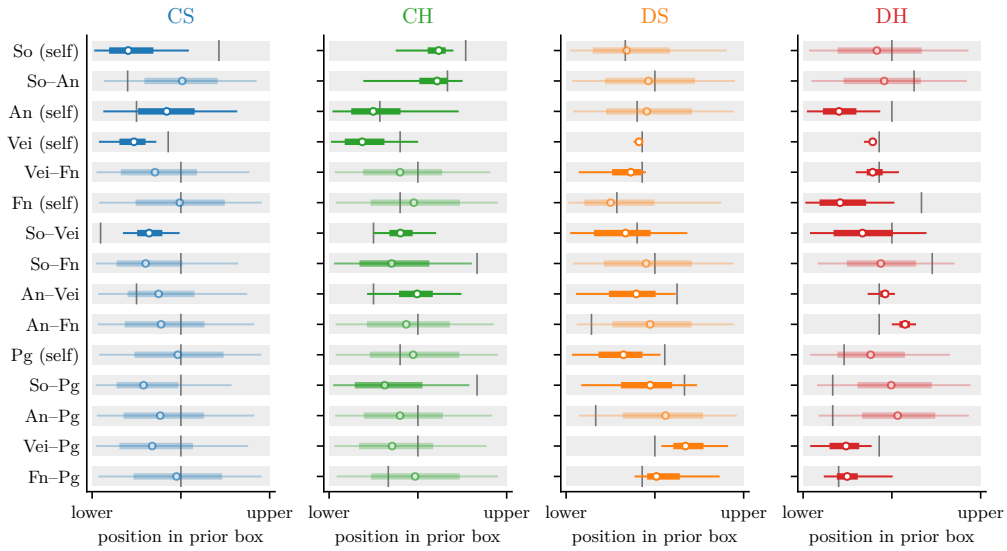


Figure 4: Marginal posteriors of the 15 entries of **A** at the final stage (three seeds pooled), drawn as position within the stage-(iii) prior box of each entry (Supplementary Table S4): thick bar 50%, thin bar 90% interval, circle median; grey tick: $a_{ij} = 0$. Dark: identified; medium: sign only; faint: not identified (as in Fig. 3). **[DS, DH from stage (iii).]**

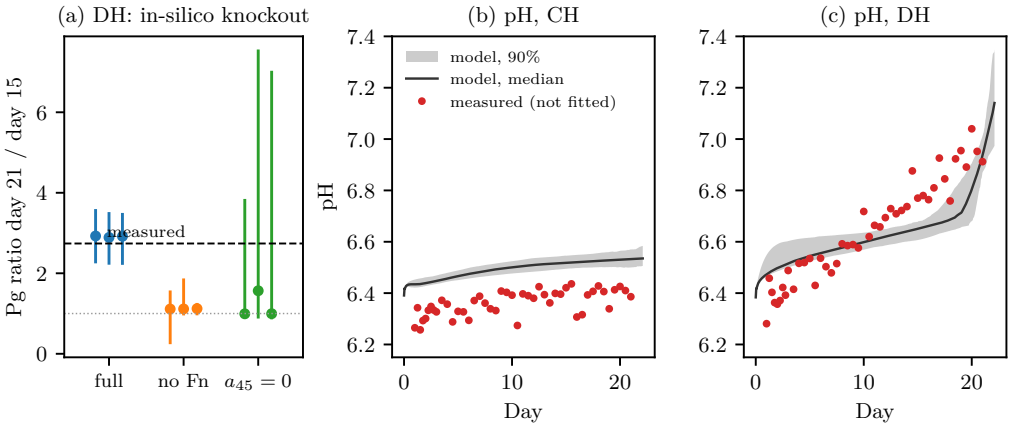


Figure 5: Checks against data and predictions not used in the inference. (a) DH: ratio of the Pg fraction on day 21 to day 15 for the calibrated model, with Fn removed, and with $a_{45} = 0$ (posterior median and 90% interval, one bar per seed; dashed: measured). (b,c) pH predicted from the posterior composition through a regression fitted once on the measured data (band: 90%), against the measured pH. **[DH from stage (iii).]**